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Read the following passage.

Azidothymidine (AZT) is a drug used to treat people infected with human immunodeficiency virus (HIV). It inhibits the enzyme that synthesises DNA from HIV RNA. This does not destroy HIV in the body but stops or slows the development of AIDS.

In the past, some people who took AZT on its own eventually developed AIDS. Some of the HIV in their bodies had become resistant to AZT. To prevent this from happening, people infected with HIV are now treated with highly active antiretroviral therapy (HAART). This involves taking AZT with other anti-HIV drugs at the same time.

5

AZT is taken in low doses. This is because people who took high doses over long periods of time suffered muscle wastage. It was found that high doses of AZT inhibit replication of mitochondria.

10

Use information from the passage and your own knowledge to answer the questions.

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1

Suggest and explain why AZT does not destroy HIV in the body but stops or slows the development of AIDS (lines 3–4).

[4 marks]

[Extra space]



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Suggest and explain **two** advantages of using HAART (lines 7–9).

[4 marks]

Advantage 1 _____

Advantage 2 _____

Question 10 continues on the next page

Turn over ►



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Suggest why high doses of AZT lead to muscle wastage (lines 10–11).

[2 marks]

10

END OF QUESTIONS**Copyright Information**

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